

# All Dice, No Slice: Metric Artifacts, Data Leakage, and Task Interference in Multi-Task Computational Pathology

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## Abstract

Simultaneous classification and lesion segmentation via hard parameter-sharing U-Nets is a compact way to couple both objectives in computational pathology. Recently, classification accuracies of 82% – 90% and Dice scores of 97% – 99% were reported across diverse medical imaging modalities using standard convolutional backbones and static loss weighting. We conduct a 26-experiment replication and methodological audit across four datasets: TCGA-LGG brain-tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the 19-tissue PanNuke benchmark. Under strict patient-level data isolation (GroupKFold), the claimed accuracies are not reproducible, and the reported Dice levels are attainable only as empty-mask artifacts: all four of our SIIM configurations converge to all-empty predictions yet report 77.74% Dice — exactly the lesion-free fraction of the validation split under the convention that scores empty-over-empty slices as perfect agreement. On 6-class ordinal prostate cancer grading (PANDA), empirical top-1 accuracy spans 29.04% – 46.15% and Dice spans 28.94% – 44.34% across all 12 evaluated configurations, with static 5:1-weighted  $10^{-3}$  baselines (Runs 03–04) reaching 45.15% – 46.15% / 43.30% – 44.34% (vs. 88.0% and 99.0% claimed). Three systemic causes explain the discrepancy: (1) empty-mask Dice crediting, (2) patient-level data leakage, and (3) task interference

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from the joint optimization recipe. Matched ablations further show that skip connections are a null result, whereas removing Macenko stain normalization *improves* subtype discrimination (+5.51% on PANDA, +2.68% on PanNuke). Our benchmark and audit protocol are released to prevent such metric artifacts in future multi-task pathology studies.

*Keywords:* Computational Pathology, Multi-Task Learning, U-Net, Reproducibility, Metric Artifacts, Data Leakage, GradNorm

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## 1. Introduction

Deep learning systems in digital pathology and radiology are increasingly required to deliver both diagnostic categorization and lesion delineation [1, 2]. Large visual foundation models (e.g., UNI [3], CONCH [4], and Virchow [5]) show that clinically actionable representations require massive pretraining and self-supervised objectives across gigapixel whole-slide images (WSIs). In resource-constrained deployments, hard parameter sharing via an encoder–decoder backbone (a U-Net [6] with dual prediction heads) remains a lightweight alternative that couples both objectives in a single forward pass.

Recently, Rhanoui et al. [7] claimed that a standard U-Net with lightweight encoders (VGG16 [8] and MobileNetV2 [9]) trained with static loss weights ( $\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$ ) and a high learning rate ( $\eta = 10^{-3}$ ) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting 82% – 90% classification accuracy and 97% – 99% Dice similarity coefficients.

In this paper, we report a 26-run replication study and systematic methodological audit. Our findings demonstrate that:

1. **Empirical Irreproducibility:** When evaluated with strict patient/biopsy-level isolation (GroupKFold), multi-class grading on PANDA prostate biopsies degrades from the claimed 88.0% accuracy / 99.0% Dice to an empirical range of 29.04% – 46.15% Acc / 28.94% – 44.34% Dice across all 12 configurations, with the static 5:1-weighted  $10^{-3}$  baselines (Runs 03–04) achieving 45.15% – 46.15% / 43.30% – 44.34%.
2. **The Empty-Mask Dice Metric Artifact:** Near-perfect (99%) Dice claims on sparse-lesion datasets such as SIIM-ACR pneumothorax can arise from crediting lesion-free slices ( $|Y| = |\hat{Y}| = 0$ ) as Dice = 1.0; our

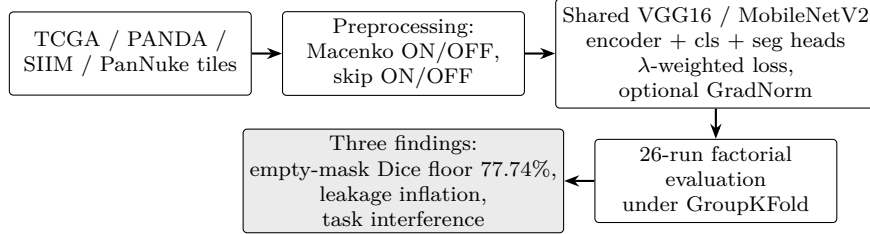


Figure 1: Overview of the 26-run replication and audit pipeline. Four source datasets (TCGA-LGG, PANDA, SIIM-ACR, PanNuke) are tiled and passed through a preprocessing stage that toggles Macenko stain normalization and skip connections. A shared VGG16/MobileNetV2 encoder feeds a classification head and a segmentation head trained with a  $\lambda$ -weighted loss (static ratios or optional GRADNORM). All 26 configurations are evaluated under patient-level GroupKFold isolation, yielding three findings: the empty-mask Dice floor of 77.74% on SIIM, patient-level leakage inflation of classification accuracy, and task interference from the joint optimization recipe.

four SIIM runs measure exactly this floor (77.74%) under an all-empty predictor.

3. **Controlled Multi-Task & Ablation Analysis:** Through matched factorial controls (learning-rate pairs, static loss-weight ratios  $\lambda_{seg} : \lambda_{cls} \in \{1:10, 1:1, 5:1, 10:1\}$ , GRADNORM [10] balancing, Macenko normalization, and skip-connection ablations), we isolate how individual components govern multi-task optimization: GradNorm-style balancing actively impairs 6-class ordinal grading (45.39%  $\rightarrow$  29.04% Acc), while skip connections show no resolvable difference.

To facilitate complete reproducibility, all training scripts, raw JSON summaries for all 26 runs, evaluation checkpoints, and generated figures are publicly available at <https://github.com/ApatheticMioz/cancer-pathology-dl> [11]. Figure 1 summarizes the end-to-end pipeline and the three findings it yields.

## 2. Multi-Task Formulation & Evaluation Metrics

### 2.1. Architectural Formulation

Let  $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$  denote an input image tile. A shared convolutional encoder  $f_\theta(\mathbf{x})$  extracts a bottleneck latent representation  $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$ . The classification branch applies global average pooling (GAP) followed by a two-layer multilayer perceptron (hidden width 256, dropout 0.5):

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}(\mathbf{W}_2 \phi(\mathbf{W}_1 \text{GAP}(\mathbf{z}) + \mathbf{b}_1) + \mathbf{b}_2) \in \Delta^{K-1}, \quad (1)$$

where  $K$  is the number of diagnostic classes,  $\Delta^{K-1}$  denotes the probability simplex, and  $\phi(\cdot)$  is the ReLU nonlinearity. The segmentation branch passes  $\mathbf{z}$  through a decoder  $\text{Dec}_\phi$  with lateral skip connections  $\{\mathbf{s}_l\}_{l=1}^L$  from encoder stages, yielding unnormalized pixel logits  $\mathbf{l}_{seg} \in \mathbb{R}^{H \times W \times M}$ . For binary segmentation ( $M = 1$ ),  $\hat{\mathbf{y}}_{seg} = \sigma(\mathbf{l}_{seg})$  with  $\sigma(\cdot)$  the sigmoid; for  $M > 1$ , class probabilities are obtained via channel-wise Softmax.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls} \mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg} \mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \quad (2)$$

where  $\mathcal{L}_{cls}$  is categorical cross-entropy  $\mathcal{L}_{CE}$ . In the published study of Rhanoui et al. [7], the segmentation loss was stated as  $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}$ ; the implementation audited for this study, however, optimizes purely cross-entropy objectives, without any Dice-based penalty term (a documented deviation):

$$\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}) = \begin{cases} \mathcal{L}_{BCE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M = 1 \text{ (binary tasks)}, \\ \mathcal{L}_{CE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M > 1 \text{ (multi-class tasks)}. \end{cases} \quad (3)$$

The Dice similarity coefficient (DSC) is evaluated post-hoc as a validation metric, not as an active loss term during gradient descent.

## 2.2. The Empty-Mask Dice Metric Artifact

The Sørensen–Dice coefficient between ground-truth mask  $Y$  and binary prediction  $\hat{Y}$  ( $Y, \hat{Y} \in \{0, 1\}^{H \times W}$ ) is:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8}; \quad \text{Dice} := 1 \text{ if } |Y| + |\hat{Y}| = 0. \quad (4)$$

Empty references and predictions are a well-recognized validation pitfall in biomedical image analysis [12]. Two conventions govern empty slices:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ( $\{i \mid |Y_i| > 0\}$ ).
- **Empty-Credit Assignment:** Assign  $\text{Dice} = 1.0$  when  $|Y| = 0 \wedge |\hat{Y}| = 0$ , and  $\text{Dice} = 0.0$  when  $|Y| = 0 \wedge |\hat{Y}| > 0$ .

Averaged over all slices under empty-credit assignment, the macroscopic mean is

$$\overline{\text{Dice}}_{\text{all}} = \frac{1}{N} \sum_{i=1}^N d_i, \quad d_i = \text{Dice}(Y_i, \hat{Y}_i) \text{ if } |Y_i| > 0, \text{ else } \mathbf{1}[|\hat{Y}_i| = 0], \quad (5)$$

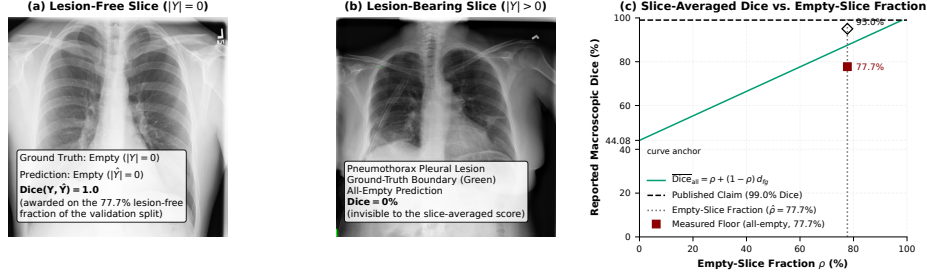


Figure 2: The empty-mask Dice artifact on SIIM-ACR pneumothorax. (a) Lesion-free slice ( $|Y| = 0$ ): under empty-credit assignment an all-empty prediction is awarded Dice = 1.0; lesion-free slices constitute  $\hat{\rho} = 77.7\%$  of our validation split. (b) Lesion-bearing slice ( $|Y| > 0$ , ground-truth boundary in green): our all-empty models predict empty masks here and score Dice = 0, invisible to the slice-averaged mean. (c) Convex-combination curve (Eq. 6): a foreground Dice of 77.74% would report 95.04% at  $\hat{\rho} = 0.777$  (analytic illustration), whereas all four of our SIIM runs measure the all-empty floor of 77.74% itself.

which for mean foreground Dice  $\overline{\text{Dice}}_{\text{fg}}$  on lesion-bearing slices is the convex combination

$$\overline{\text{Dice}}_{\text{all}} = \rho + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \quad (6)$$

where  $\rho$  is the proportion of lesion-free slices. In sparse pathologies such as SIIM-ACR pneumothorax, a model with a genuine foreground Dice of 77.74% would report  $\overline{\text{Dice}}_{\text{all}} = 0.777 \times 1.0 + 0.223 \times 0.7774 = 95.04\%$  at the observed lesion-free fraction  $\hat{\rho} = 1,659/2,135 = 0.777$  (an analytic illustration). Our four SIIM runs realize a more extreme failure mode. They emit empty masks on every validation slice throughout training, and each reported 77.74% (Runs 05, 06, 11, 12) *is the all-empty floor itself* — Eq. 5 with zero foreground Dice. Fig. 2 shows why this matters: on sparse data, slice-averaged Dice reports the lesion-free fraction rather than lesion overlap. Published 99.0% Dice claims [7] are therefore uninterpretable as evidence of lesion localization.

### 2.3. Adaptive Gradient Balancing via GradNorm

To test whether dynamic loss balancing mitigates cross-task gradient competition, we introduced a parameterized variant of GRADNORM [10] in Phase v2.

Let  $W$  denote the shared convolutional filters and normalization layers of the U-Net encoder  $f_{\theta}$ . The  $L_2$  gradient norm of task  $i \in \{\text{seg}, \text{cls}\}$  with

respect to  $W$  at minibatch  $t$  is:

$$G_W^{(i)}(t) = \|\nabla_W w_i(t) \mathcal{L}_i(t)\|_2. \quad (7)$$

For stability in the primary **Adam** optimizer, balancing weights are learnable log-weights  $\mu_i(t) = \ln w_i(t)$  (initialized to  $w_{seg}(0) = 5.0$ ,  $w_{cls}(0) = 1.0$ ), updated jointly with the network parameters. Minibatch optimization minimizes the  $L_1$  gradient disparity objective:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg, cls\}} \left| G_W^{(i)}(t) - \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right|, \quad (8)$$

where  $\bar{G}_W(t) = \frac{1}{2} \sum_j G_W^{(j)}(t)$  is the average gradient norm across tasks,  $\alpha = 1.5$  is the asymmetry parameter, and  $r_i(t)$  represents the relative inverse training rate:

$$r_i(t) = \frac{\tilde{\mathcal{L}}_i(t)}{\frac{1}{2} \sum_{j \in \{seg, cls\}} \tilde{\mathcal{L}}_j(t)}, \quad \text{with} \quad \tilde{\mathcal{L}}_i(t) = \frac{\mathcal{L}_i(t)}{\mathcal{L}_i(0)}, \quad (9)$$

where  $\mathcal{L}_i(0)$  is the initial step-0 loss. Following Chen et al. [10], slower-training tasks exhibit higher loss ratios  $r_i(t) > 1$ , which raises their target gradient norm  $\bar{G}_W(t)[r_i(t)]^\alpha$ . After each optimizer update, we renormalize the weights so that  $\sum_i w_i(t) = 2$ . In the canonical formulation [10], the module differentiates  $\mathcal{L}_{grad}$  against detached targets and applies it to the balancing weights alone through a dedicated optimizer (learning rate 0.025). §4.3 probes this distinction empirically. We evaluate the module across Groups 2 and 4 to isolate how dynamic gradient equalization behaves in hard-parameter-sharing architectures.

#### 2.4. Color Deconvolution and Optical Density Space

To mitigate staining variability across clinical laboratories, the v2 pipeline incorporates Macenko stain normalization [13]. We convert RGB pixel intensities  $\mathbf{I} \in [0, 255]^3$  to optical density (OD) space:

$$\mathbf{OD} = -\ln \left( \frac{\mathbf{I} + \epsilon}{I_0} \right), \quad (10)$$

where  $I_0 = 255$  represents transmitted light intensity. The implementation computes optical density via natural logarithms  $-\ln(\cdot)$  rather than the canonical base-10 formulation  $-\log_{10}(\cdot)$  [13]. We estimate the hematoxylin and

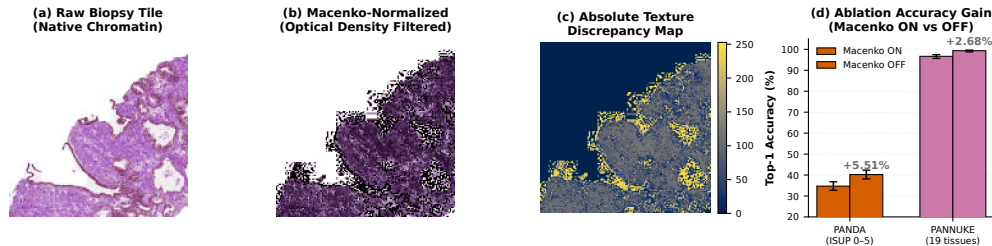


Figure 3: Visual and quantitative impact of Macenko stain normalization on histopathology classification. (a) Raw prostate biopsy tile. (b) Optical-density-projected tile: color deconvolution acts as a chrominance low-pass filter, flattening sub-cellular texture. (c) Absolute texture discrepancy map. (d) Empirical top-1 accuracy gains across matched models when Macenko normalization is bypassed (+5.51% on PANDA, +2.68% on PanNuke).

eosin stain vectors by eigendecomposition of the optical-density covariance matrix (stain-vector angles restricted to the 1st–99th percentile range) over OD values above a threshold of 0.15, and project them onto a target reference vector. This step effectively eliminates macroscopic batch variations. It also acts as a chrominance low-pass filter, however: projecting raw RGB channels onto static stain vectors smooths the sub-cellular chromatin textures that are clinically essential for grading fine glandular architectures (Fig. 3).

### 3. The 26-Run Empirical Reproduction Campaign

#### 3.1. Datasets and Experimental Protocol

We evaluated four benchmark datasets:

1. **TCGA-LGG Brain Tumor:** 2-class lesion classification and FLAIR abnormality segmentation on 2D  $256 \times 256$  slices ( $N = 3,929$ ; 110 patients [14]).
2. **PANDA Prostate Biopsies:** 6-class ISUP Gleason grading and gland segmentation ( $N = 10,616$  tiles [15]; patient-level GroupKFold isolation).
3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ( $N = 12,047$  images [16]).
4. **PanNuke:** 19-tissue multi-organ benchmark ( $N = 7,901$  patches [17]); having no published multi-task targets, it serves as an unclaimed external control.

We trained all models with PyTorch (up to 50 epochs with early stopping, patience 10; batch size 32; fixed seed 42) using ImageNet-pretrained encoders and the Adam optimizer with a constant per-phase learning rate and zero weight decay, and we applied no learning-rate schedule or additional regularization. In Phase v1 (Runs 01–06), we replicated the original static setup ( $\eta = 10^{-3}$ ,  $\lambda_{seg} : \lambda_{cls} = 5 : 1$ , raw inputs). In Phase v2 (Runs 07–12), we incorporated dynamic GRADNORM ( $\alpha = 1.5$ ,  $\eta = 10^{-4}$ ) and Macenko stain normalization [13]. Groups 3–5 (Runs 13–26) systematically evaluate multi-organ generalizability, optimization hyperparameter isolation, and architectural skip ablations. Table 1 summarizes all 26 runs and Figure 4 visualizes them.

## 4. Discussion & Methodological Anatomy

### 4.1. The Anatomy of the PANDA Collapse

In Table 1, the most substantial discrepancy occurs on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class ISUP grading alongside a 99.0% Dice score. In contrast, across all 12 PANDA configurations tested, empirical top-1 accuracy spans 29.04% – 46.15% and Dice spans 28.94% – 44.34%. The static 5:1-weighted  $10^{-3}$  baselines (Runs 03–04) achieve 45.15% / 43.30% (VGG16) and 46.15% / 44.34% (MobileNetV2). Run 04 is the strongest single configuration in our campaign.

This outcome is consistent with established clinical and machine learning benchmarks. In the international PANDA challenge (1,010 participating teams) [15], the top-performing whole-slide multi-instance learning algorithms reached external validation quadratic weighted  $\kappa$  of 0.862 (95% CI 0.840–0.884) and 0.868 (0.835–0.900) on U.S. and European cohorts, and inter-observer Gleason grading of biopsy cores is itself imperfect ( $\kappa = 0.68$  among urologic pathologists and 0.435 among general pathologists [18, 19]). Claiming 88.0% accuracy on  $128 \times 128$  isolated patches with a standard 2D CNN is not attainable without patient-level data leakage: tiles from the same biopsy core contaminate both train and test splits. Figure 5 traces the PANDA accuracy drop across training epochs and isolates the responsible mechanism.

### 4.2. Controlled Skip Connection Ablation: A Verified Null Result

Skip connections are evaluated in matched pairs (learning rate, Grad-Norm, Macenko normalization held identical):

- **Coarse macro-tumor structures (TCGA):** Run 25 (skip-ablated) vs. matched baseline Run 08 (both  $\eta = 10^{-4}$ , GradNorm, Macenko): accuracy 93.32%  $\rightarrow$  94.34% ( $\Delta_{\text{Acc}} = +1.02$ ), Dice 84.20% vs 84.12% ( $\Delta_{\text{Dice}} = -0.08$ ).
- **Fine micro-glandular structures (PANDA):** Run 10 (34.70% Acc / 31.34% Dice) vs. matched skip-ablated Run 26 (35.31% / 28.94%):  $\Delta_{\text{Acc}} = +0.61$ ,  $\Delta_{\text{Dice}} = -2.40$ .

In both cases the differences are small, and the 95% Wilson score intervals on top-1 accuracy overlap substantially (TCGA: [91.34, 94.87] vs [92.49, 95.76]; PANDA: [32.69, 36.76] vs [33.30, 37.38]). We therefore report these as *verified null results*: the bottleneck-only representation suffices for both coarse macro-lesion and fine micro-glandular segmentation; zeroing the skips produces no detectable penalty in a single-run comparison. A definitive directional claim would require multi-seed replication of the matched pairs.

*Macenko stain-normalization ablation..* The matched Macenko pair (Group 5, Runs 23–24) is the only single-run ablation with a statistically resolvable classification shift: omitting normalization raises PANDA top-1 accuracy by +5.51 and PanNuke accuracy by +2.68 (disjoint 95% CIs), while Dice stays flat on PANDA and rises by +12.79 on PanNuke. Color deconvolution thus harms fine-grained classification, consistent with over-smoothing of the chromatin/texture cues on which grading relies, and partially decomposes the v1-vs-v2 PANDA gap of §4.1. Figure 6 consolidates all six matched pairs; Figure 3 shows the texture flattening.

#### 4.3. Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA  $\times$  VGG16) factorizes the key optimization hyperparameters: learning rate ( $\eta \in \{10^{-3}, 10^{-4}\}$ ), GradNorm-style balancing (on/off), and static loss-weighting ratios  $\lambda_{\text{seg}} : \lambda_{\text{cls}} \in \{1:10, 1:1, 5:1, 10:1\}$ , all under raw inputs.

*Direct Isolation of Dynamic Gradient Balancing (Run 18)..* The isolated GradNorm arm at  $\eta = 10^{-3}$  under matched raw-input conditions (Run 18: PANDA  $\times$  VGG16, GradNorm  $\alpha = 1.5$ , raw input) compares directly against the static 5:1 baseline (Run 20). Activating GradNorm ( $\alpha = 1.5$ ) causes validation accuracy to collapse from 45.39% down to 29.04% (95% Wilson CI: [27.14, 31.02]), alongside a corresponding drop in segmentation Dice from

44.08% down to 31.43% ( $\Delta_{\text{Acc}} = -16.35\%$ ,  $\Delta_{\text{Dice}} = -12.65\%$ , with non-overlapping confidence intervals: [43.27, 47.52] vs [27.14, 31.02]).

This 29.04% accuracy and 31.43% Dice matches the performance observed in Phase v2 at  $\eta = 10^{-4}$  (Run 09: 30.89% Acc / 31.59% Dice). The  $\approx 30\%$  floor on 6-class PANDA grading is thus driven primarily by the  $\alpha = 1.5$  balancing formulation, whose gradient-norm equalization forces destructive updates onto the shared encoder, rather than by the lower learning rate.

*Learning Rate and Loss Weighting Dynamics..* Holding GradNorm off, reducing the learning rate from  $\eta = 10^{-3}$  (Run 20: 45.39% Acc) to  $\eta = 10^{-4}$  (Run 17: 43.35%) yields a minor change within overlapping 95% CIs. The static loss-ratio sweep (Runs 19–22) confirms  $\lambda = 5:1$  as the best static balance, yet no ratio approaches the claimed 88% (Figure 7). The isolated GradNorm arm (Run 18) drops to 29.04% against the 45.39% static baseline with non-overlapping CIs, and the canonical decoupled scheme plateaus at 43.06% — dynamic balancing in either formulation cannot bridge the gap (Figures 5, 6).

## 5. Conclusion & Recommendations

Across 26 experiments, the claimed 88% – 99% metrics did not reproduce under patient isolation. The residual gap stems from dataset difficulty and the joint optimization recipe: skip connections are a null result, while joint balancing ( $\alpha = 1.5$ ) destabilizes classification (45.39%  $\rightarrow$  29.04% Acc). We recommend: (1) mandatory patient/biopsy-level GroupKFold splitting; (2) explicit reporting of the Dice evaluation convention, with foreground-only or zero-penalty scoring on sparse-lesion datasets; and (3) 95% Wilson score intervals on matched single-run ablation comparisons to distinguish genuine architectural effects from stochastic variance. We further recommend (4) validation on diverse multi-organ benchmarks such as PanNuke, and (5) systematic evaluation of multi-task balancing hyperparameters before clinical translation.

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Table 1: 26-run experimental results matrix across all five evaluation groups. V1 = original static baseline ( $\eta = 10^{-3}$ ,  $\lambda_{seg} : \lambda_{cls} = 5 : 1$ , raw input); V2 = GradNorm variant ( $\alpha = 1.5$ ,  $\eta = 10^{-4}$ , Macenko normalization).  $\checkmark$  = GradNorm active;  $-$  = not used. Markers:  $\times$  = skip connections zeroed;  $\diamond$  = raw input without Macenko normalization (v1 runs are raw by construction). Rows 17–22 (Group 4) systematically isolate learning rate ( $\eta = 10^{-4}$  vs  $10^{-3}$ ), GradNorm ( $\alpha = 1.5$ , Run 18), and loss weighting ratios on PANDA  $\times$  VGG16 under raw input. Pub. Acc/Dice: published reference targets of Rhanoui et al. [7] ( $-$  = no published target).  $\Delta_{Acc}$  /  $\Delta_{Dice}$ : difference relative to published targets. All metrics in %. Complete 95% Wilson score intervals for all 26 configurations are tabulated in the open-source repository matrix (paper\_results\_matrix\_with\_ci.csv).

| #                                                                                                                    | Configuration                                  | Dataset | Encoder     | LR        | $\eta$       | GN           | Val   | Acc   | Val  | Dice | Pub. | Acc  | Pub. | Dice | $\Delta_{Acc}$ | $\Delta_{Dice}$ |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------|-------------|-----------|--------------|--------------|-------|-------|------|------|------|------|------|------|----------------|-----------------|
| <b>Group 1: Baseline Reproductions (static 5:1, <math>\eta = 10^{-3}</math>, raw input)</b>                          |                                                |         |             |           |              |              |       |       |      |      |      |      |      |      |                |                 |
| 01                                                                                                                   | Naked Baseline                                 | TCGA    | VGG16       | $10^{-3}$ | —            | —            | 88.82 | 76.02 | 89.0 | 97.0 | —    | 97.0 | —    | —    | −0.18          | −20.98          |
| 02                                                                                                                   | Naked Baseline                                 | TCGA    | MobileNetV2 | $10^{-3}$ | —            | —            | 94.60 | 87.56 | 90.0 | 98.0 | —    | 98.0 | —    | —    | +4.60          | −10.44          |
| 03                                                                                                                   | Naked Baseline                                 | PANDA   | VGG16       | $10^{-3}$ | —            | —            | 45.15 | 43.30 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −41.85         | −54.70          |
| 04                                                                                                                   | Naked Baseline                                 | PANDA   | MobileNetV2 | $10^{-3}$ | —            | —            | 46.15 | 44.34 | 88.0 | 99.0 | —    | 99.0 | —    | —    | −41.85         | −54.66          |
| 05                                                                                                                   | Naked Baseline                                 | SIIM    | VGG16       | $10^{-3}$ | —            | —            | 83.19 | 77.74 | 82.0 | 99.0 | —    | 99.0 | —    | —    | +1.19          | −21.26          |
| 06                                                                                                                   | Naked Baseline                                 | SIIM    | MobileNetV2 | $10^{-3}$ | —            | —            | 83.47 | 77.74 | 87.0 | 99.0 | —    | 99.0 | —    | —    | −3.53          | −21.26          |
| <b>Group 2: V2 Package (GradNorm <math>\alpha = 1.5</math>, <math>\eta = 10^{-4}</math>, Macenko normalization)</b>  |                                                |         |             |           |              |              |       |       |      |      |      |      |      |      |                |                 |
| 07                                                                                                                   | GradNorm v2                                    | TCGA    | VGG16       | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 92.80 | 78.30 | 89.0 | 97.0 | —    | 97.0 | —    | —    | +3.80          | −18.70          |
| 08                                                                                                                   | GradNorm v2                                    | TCGA    | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 93.32 | 84.20 | 90.0 | 98.0 | —    | 98.0 | —    | —    | +3.32          | −13.80          |
| 09                                                                                                                   | GradNorm v2                                    | PANDA   | VGG16       | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 30.89 | 31.59 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −56.11         | −66.41          |
| 10                                                                                                                   | GradNorm v2                                    | PANDA   | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 34.70 | 31.34 | 88.0 | 99.0 | —    | 99.0 | —    | —    | −53.30         | −67.66          |
| 11                                                                                                                   | GradNorm v2                                    | SIIM    | VGG16       | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 81.08 | 77.74 | 82.0 | 99.0 | —    | 99.0 | —    | —    | −0.92          | −21.26          |
| 12                                                                                                                   | GradNorm v2                                    | SIIM    | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 83.00 | 77.74 | 87.0 | 99.0 | —    | 99.0 | —    | —    | −4.00          | −21.26          |
| <b>Group 3: PanNuke 19-Tissue Multi-Organ Benchmark (external control; no published targets)</b>                     |                                                |         |             |           |              |              |       |       |      |      |      |      |      |      |                |                 |
| 13                                                                                                                   | Naked Baseline                                 | PanNuke | VGG16       | $10^{-3}$ | —            | —            | 98.85 | 72.48 | —    | —    | —    | —    | —    | —    | —              | —               |
| 14                                                                                                                   | Naked Baseline                                 | PanNuke | MobileNetV2 | $10^{-3}$ | —            | —            | 99.43 | 74.21 | —    | —    | —    | —    | —    | —    | —              | —               |
| 15                                                                                                                   | GradNorm v2                                    | PanNuke | VGG16       | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 91.90 | 39.59 | —    | —    | —    | —    | —    | —    | —              | —               |
| 16                                                                                                                   | GradNorm v2                                    | PanNuke | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 96.68 | 60.54 | —    | —    | —    | —    | —    | —    | —              | —               |
| <b>Group 4: Optimization Isolation (PANDA <math>\times</math> VGG16: LR, GradNorm, loss weighting; raw input)</b>    |                                                |         |             |           |              |              |       |       |      |      |      |      |      |      |                |                 |
| 17                                                                                                                   | Isolate LR: static $\eta = 10^{-4}$ $\diamond$ | PANDA   | VGG16       | $10^{-4}$ | —            | —            | 43.35 | 37.99 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −43.65         | −60.01          |
| 18                                                                                                                   | Isolate GradNorm: $\alpha = 1.5$               | PANDA   | VGG16       | $10^{-3}$ | $\checkmark$ | $\checkmark$ | 29.04 | 31.43 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −57.96         | −66.57          |
| 19                                                                                                                   | Loss weight $\lambda = 1:1$                    | PANDA   | VGG16       | $10^{-3}$ | —            | —            | 42.73 | 41.72 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −44.27         | −56.28          |
| 20                                                                                                                   | Loss weight $\lambda = 5:1$                    | PANDA   | VGG16       | $10^{-3}$ | —            | —            | 45.39 | 44.08 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −41.61         | −53.92          |
| 21                                                                                                                   | Loss weight $\lambda = 1:10$                   | PANDA   | VGG16       | $10^{-3}$ | —            | —            | 41.83 | 38.06 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −45.17         | −59.94          |
| 22                                                                                                                   | Loss weight $\lambda = 10:1$                   | PANDA   | VGG16       | $10^{-3}$ | —            | —            | 44.58 | 41.52 | 87.0 | 98.0 | —    | 98.0 | —    | —    | −42.42         | −56.48          |
| <b>Group 5: Ablation Matrix (stain normalization <math>\diamond</math> and skip connections <math>\times</math>)</b> |                                                |         |             |           |              |              |       |       |      |      |      |      |      |      |                |                 |
| 23                                                                                                                   | Ablation: No-Macenko $\diamond$                | PANDA   | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 40.21 | 31.57 | 88.0 | 99.0 | —    | 99.0 | —    | —    | −47.79         | −67.43          |
| 24                                                                                                                   | Ablation: No-Macenko $\diamond$                | PanNuke | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 99.36 | 73.33 | —    | —    | —    | —    | —    | —    | —              | —               |
| 25                                                                                                                   | Ablation: No-Skips $\times$                    | TCGA    | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 94.34 | 84.12 | 90.0 | 98.0 | —    | 98.0 | —    | —    | +4.34          | −13.88          |
| 26                                                                                                                   | Ablation: No-Skips $\times$                    | PANDA   | MobileNetV2 | $10^{-4}$ | $\checkmark$ | $\checkmark$ | 35.31 | 28.94 | 88.0 | 99.0 | —    | 99.0 | —    | —    | −52.69         | −70.06          |

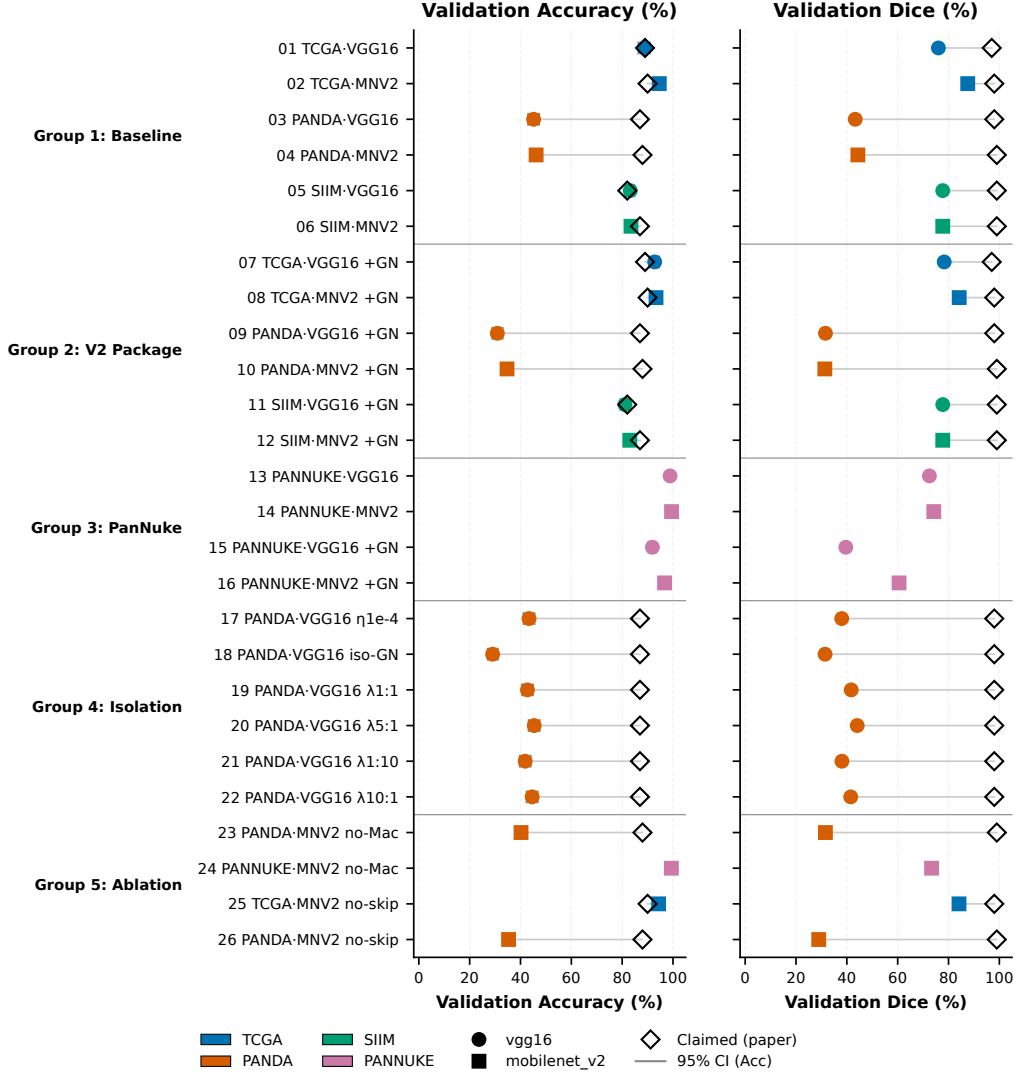


Figure 4: 26-run claimed-vs-measured dot plot. Left panel: validation accuracy (%) with 95% Wilson-CI whiskers (measured, filled markers) and the published claimed accuracy (black open diamond); the tie line between them is the gap that constitutes the paper’s central argument. Right panel: validation Dice (%) with the same layout but no whiskers, as no published confidence intervals exist for the macro-Dice claims. The PANDA rows (Runs 03–04, 09–10, 17–22, 23, 26) show the largest gaps: measured accuracy spans 29.04%–46.15% vs. claimed 87–88%, and Dice spans 28.94%–44.34% vs. claimed 98–99%. All four SIIM rows (Runs 05, 06, 11, 12) measure exactly 77.74% Dice — the all-empty floor. PanNuke rows (Runs 13–16, 24) have no published targets and are shown without claimed diamonds.

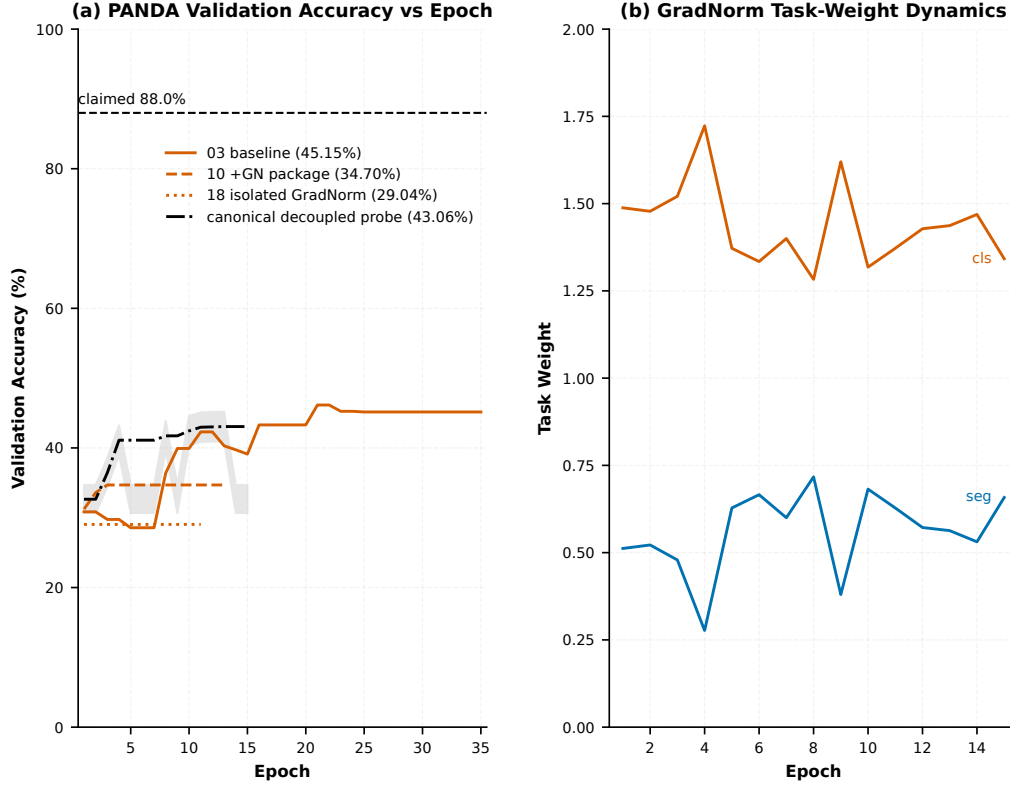


Figure 5: Anatomy of the PANDA accuracy drop. (a) Validation accuracy vs. epoch for three PANDA runs — Run 03 (static 5:1 baseline, 45.15%), Run 10 (v2 GradNorm package, 34.70%), Run 18 (isolated GradNorm at  $\eta = 10^{-3}$ , 29.04%) — plus the canonical decoupled GRADNORM probe (dash-dot, shaded 95% CI band), which remains stable but plateaus at 43.06% over 15 epochs, 44.94 percentage points below the claimed 88.0% (black dashed reference). (b) GRADNORM task-weight dynamics from the canonical log: the segmentation and classification weights over 15 epochs, showing the dynamic rebalancing that the joint variant turns into destructive updates on the shared encoder.

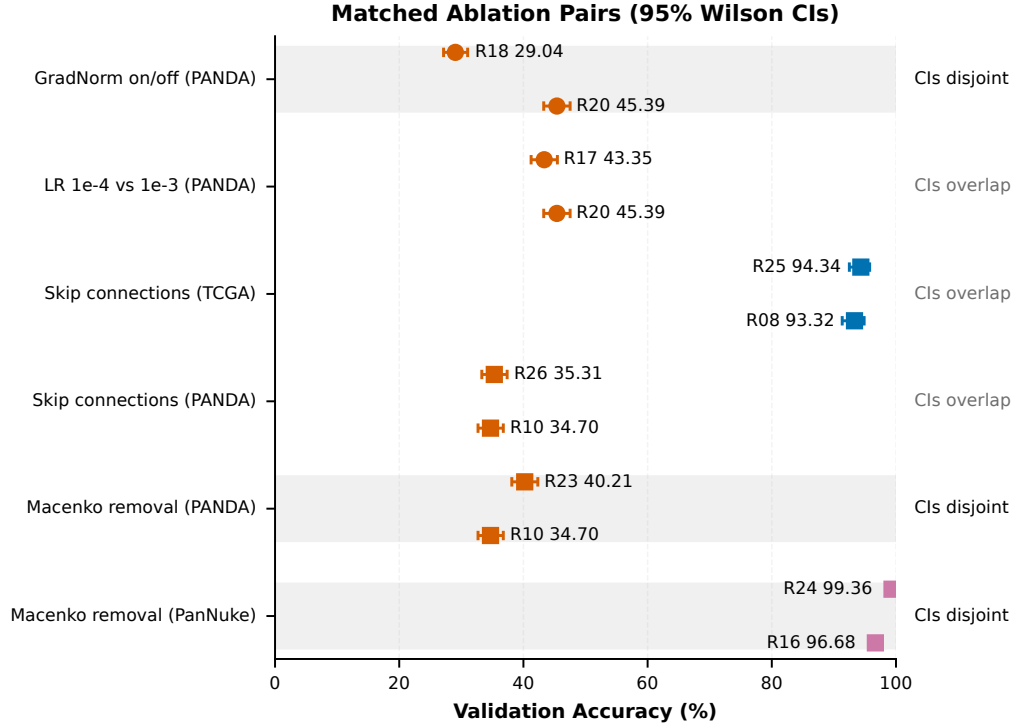


Figure 6: Matched-ablation forest plot of validation accuracy with 95% Wilson CIs for the six strictly matched pairs. Disjoint-CI pairs (shaded): isolated GradNorm (Run 18, 29.04%) vs. static 5:1 (Run 20, 45.39%); no-Macenko PANDA (Run 23, 40.21%) vs. matched (Run 10, 34.70%); no-Macenko PanNuke (Run 24, 99.36%) vs. matched (Run 16, 96.68%). Overlapping-CI pairs: LR  $10^{-4}$  vs.  $10^{-3}$  (Runs 17 vs. 20: 43.35% vs. 45.39%); skip ablation TCGA (Runs 25 vs. 08: 94.34% vs. 93.32%); skip ablation PANDA (Runs 26 vs. 10: 35.31% vs. 34.70%). The right-margin verdict column encodes the CI-overlap test, so the null-result status of the skip ablations and the resolvable Macenko/GradNorm effects are read directly from the plot.

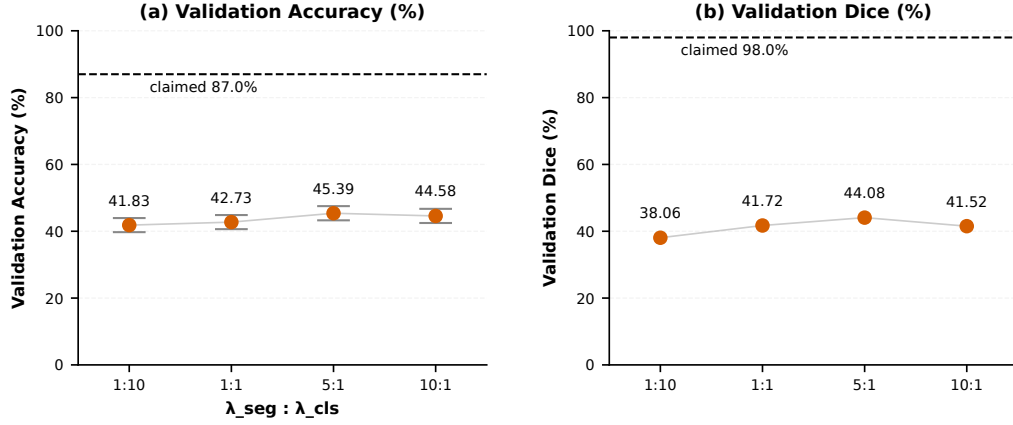


Figure 7: Static loss-weight ratio sweep on PANDA  $\times$  VGG16 (raw input,  $\eta = 10^{-3}$ , no GradNorm). (a) Validation accuracy with 95% Wilson-CI whiskers:  $\lambda_{seg} : \lambda_{cls} = 1:10$  (41.83%), 1:1 (42.73%), 5:1 (45.39%), 10:1 (44.58%); the black dashed line marks the claimed 87.0% accuracy. (b) Validation Dice for the same four arms (38.06%, 41.72%, 44.08%, 41.52%) against the claimed 98.0%. The 5:1 ratio is the best static balance on this architecture, yet no ratio approaches the claimed target — the gap is not a loss-weighting artifact.